

CHEN SHAO

cshao676@gmail.com

EDUCATION

PhD Candidate in Computer Science Karlsruhe Institute of Technology (KIT), Germany <i>Supervised by Prof. York Sure-Vetter (KIT-AIFB) and Prof. Danaï Koutra (UMich)</i>	July 2022 – Present
M.Sc. in Electronic and Information Engineering Karlsruhe Institute of Technology (KIT), Germany <i>Thesis: “Energy Difference Prediction in Chemistry Using Graph Neural Networks”</i> <i>Supervised by Prof. Pascal Friederich (AiMat, ITI)</i>	Oct. 2018 – Mar. 2022 Grade: 2.1
B.Eng. in Electronic and Information Engineering Xi'an Jiaotong University (XJTU), China <i>Thesis: Ground-Roll Noise Removal Based on Multiscale Geometric Transformation</i> <i>Honors: Siyuan Scholarship (Top 5% of class)</i> <i>Awards: First Prize, National Undergraduate Innovation and Entrepreneurship Competition</i>	Sept. 2013 – June 2017 Grade: 1.8

TECHNICAL EXPERTISE

NLP: Fine-tuning and pre-training of language models, multimodal alignment.
AI/ML Architecture: Generative Models (Flow Matching/Diffusion), Graph Foundation Models.
Software & Engineering: Python (PyTorch, PyG, JAX, TensorFlow), C/C++, Docker, MATLAB.
Languages: Chinese (Native), English (Business Fluent), German (C1).

PROJECTS

Research Intern, Max Planck Institute of Biochemistry <i>Foundation Models for Protein Design</i> Supervised by Dr. Dexiong Chen and Prof. Karsten Borgwardt	01/2026 – Present Munich, Germany
Developed and optimized pre-training algorithms for foundation models in molecular property prediction.	
Research Associate (Research, PI), Karlsruhe Institute of Technology <i>Graphs in Time Series and Natural Sciences</i>	06/2022 – Present Karlsruhe, Germany
LLM-GALP (LLM Graph Reasoning (PI), Funder BMFTR)	09/2024 – Present
Developed SOTA memory-efficient LLM fine-tuning for text-attributed graphs and submitted to PKDD (Top Journal in Data Mining).	
ML4RW (Responsible AI for Energy System (PI), Funder KIT-ARRTI)	01/2024 – 01/2025
Designed a state-of-the-art approach for time series forecasting and provided a new benchmark of European energy dataset for large-scale, non-stationary time-series research.	
KIGLIS (AI in 6G with Nokia Bell Lab, VPI Photonics, Funder BMFTR)	06/2022 – 01/2024
- Actively contributed in a collaboration project <i>Nokia Bell Labs</i> and <i>VPI Photonics</i>. - Proposed two novel algorithms for 6G channel equalization and presented them in ECOC and OFC.	
Research Assistant (Hiwi), Karlsruhe Institute of Technology (KIT) <i>Different Engagements in signal processing and machine learning</i>	11/2018 – 06/2022 Karlsruhe, Germany
Autonomous Learning Robots (ALR Lab), KIT <i>Hiwi Student: Graph Neural Networks for Finite Element Analysis</i> Supervised by Niklas Freymuth (KIT)	11/2021 – 04/2022

- Generated a synthetic Finite Element Simulation dataset to train an adaptive meshing algorithm, supporting reinforcement learning environments for high-resolution simulators.

Institute of Industrial Information Technology (IIIT), KIT

06/2020 – 04/2021

Hiwi Student: Algorithm Development for Hyperspectral Imaging

Supervised by Dr. Maximilian Schambach (KIT, SAP)

- Developed software to systematically process light field data using time-frequency decomposition.

Precitec GmbH & Co. KG

04/2021 – 09/2021

Industry Research Intern, Deep Learning for Laser Material Processing

Gaggenau/Karlsruhe, Germany

- Benchmarked five deep learning architectures for image classification and optimized performance for laser material processing video sequences.

Xi'an Jiaotong University (XJTU)

01/2016 – 06/2017

Research Intern, Ground-roll Noise Removal Based on Multiscale Geometric Transformation

Xi'an, China

Supervised by Prof. Wenchao Chen (XJTU)

- Developed and adapted Ridgelet algorithms for noise suppression in raw seismic data.

PRINCIPAL INVESTIGATOR IN FUNDED PROJECTS

(ML4RW) Machine Learning-Based Algorithm for Maximizing Resource Waste Reduction.

Funded by *KIT Academy for Responsible Research, Teaching, and Innovation (ARRTI)*, 2024.

EUR 74,000

(LLM-GALP) Large Language Model Featured Graph Algorithms for Link Prediction.

Funded by *Federal Ministry of Education and Research (BMFTR)*, 2025–2026.

EUR 105,800

AWARDS AND FELLOWSHIPS

Software Campus Fellowship – “Werde EntrepreNerd”

2025 – 2026

Awarded for outstanding publication record and leadership potential.

Siyuan Scholarship (Top 5%)

2016

Academic excellence award at Xi'an Jiaotong University.

National Innovation Competition (First Prize)

2017

Recognized for independent cutting edge research at XJTU.

Provincial String Competition (First Prize)

2011

First place in a province-level high school violin competition.

SELECTED PUBLICATIONS & AWARDS

Nature Communications Materials (2022): “Review of Graph Neural Network for Materials”.

TKDD/KDD (2025): Lead author on Graph Automorphism & Link Prediction research.

Awards: DLR Fellowship (2025), Siyuan Scholarship, National Innovation Competition Winner.

PEER-REVIEWED PUBLICATIONS

Zhenyi Zhu, **Chen Shao**, et al. *Deep Switching Koopman Networks: Discovery of Switching Dynamic Regimes in Industrial Systems. Proceedings of the 2026 IEEE 22nd International Conference on Automation Science and Engineering (CASE 2026)*, Shenyang, China, August 17–21, 2026.

Chen Shao, Zhenyi Zhu, and Danai Koutra. *DynGCN: Capturing Dynamic Correlation with Message Passings*. In *ICML 2026 Workshop on Foundation Models for Structured Data (FMSD)*, 2026.

Chen Shao, Joella Wang, Zhanbo Huang, Danai Koutra. *When Forecasting Fails: Overcoming Temporal Correlation Shift in Time Series*. In *European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML PKDD)*, 2026.

Chen Shao, Joella Wang, Zhanbo Huang, Tobias Kaefer, Michael Faerber, Benjamin Schaefer. *Real-E: A Foundation Benchmark for Advancing Robust and Generalizable Electricity Forecasting*. *Proceedings of the ACM International Conference on Information and Knowledge Management (CIKM)*, 2025.

Chen Shao, Ruirui Wang, Tobias Kaefer, Danai Koutra. *Bridging Semantics and Structure: A Theoretically Grounded Framework for Link Prediction on TAGs*. *ACM Transactions on Knowledge Discovery from Data (TKDD)*, 2026.

Chen Shao, Elias Giacomidis, Shi Li, Jialei Li, Michael Faerber, Tobias Kaefer, and Andre Richter. *A Novel Machine Learning-Based Equalizer for a Downstream 100G PAM-4 PON*. *Optical Fiber Communication Conference and Exhibition (OFC 2024)*, Oral Presentation (Top 5%).

Chen Shao, Elias Giacomidis, Tobias Kaefer, Michael Faerber. *Time-Series Forecasting Methods in Fiber Nonlinearity Mitigation*. *European Conference on Optical Communication (ECOC 2024)*, Oral Presentation (Top 5%).

Chen Shao, Elias Giacomidis, Michael Faerber, Tobias Kaefer. *Time-Series Prediction Methods in Fiber Nonlinearity Mitigation*. *Photonics*, 2024. Special Issue: Machine Learning Applied to Optical Communication Systems (Invited Paper).

Patrick Reiser, **Chen Shao**, Chen Zhou, Andre Eberhard, Houssam Metni, Marlen Neubert, Timo Sommer, Clint van Hoesel, Luca Torresi, Pascal Friederich. *Review of Graph Neural Networks for Materials*. *Communications Materials*, 3, 93 (2022).

PREPRINTS

Chen Shao, Donald Loveland, Danai Koutra. *Edge-Level Automorphism in GNNs: A Quantitative Framework and Effective Designs for Link Prediction*. (Submitted to ICLR27).

Chen Shao, Chen Zhou, Pascal Friederich. *Graph Neural Networks to Learn Joint Representations of Disjoint Molecular Graphs*. *arXiv:2210.09517* (2022).

REFERENCES

Dr. Tobias Käfer

Senior Researcher / Group Leader
Karlsruhe Institute of Technology (KIT)
Institute AIFB
Email: tobias.kaefer@kit.edu
Relationship: Doctoral Supervisor

Prof. Danai Koutra

Associate Professor of Data Science and AI
University of Michigan, Ann Arbor
Computer Science and Engineering
Email: dkoutra@umich.edu
Relationship: Co-Supervisor